A2 More Notes on 8-3C **The tricky stuff:**

What if you have a hole at the same place that you have an x intercept?

⇒ The hole wins! Make sure you don't leave it as an eyeball ⊙ Make it just a hole, no x intercept

What if you have a Vertical Asymptote at the same place you have a hole?

⇒ you have a VA which is more than just a hole V.A. wins!

What if you have no features to graph on one side of the y-axis?

⇒ Pick a value for x that is ~~...~~ on the side in question and find it's y value. wherever that (x,y) lands lets you know where to draw the brance of the graph

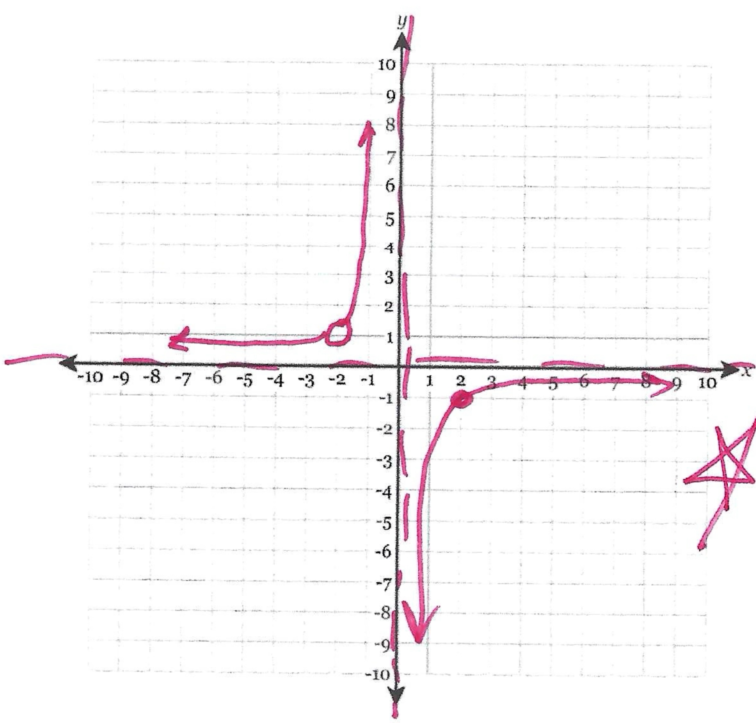
You! 7. Identify all vertical and horizontal asymptotes, x- and y-intercepts, and holes, and then use them to graph the rational function below.

$$f(x) = \frac{-2x-4}{x^2+2x} = \frac{-2\cancel{(x+2)}}{x\cancel{(x+2)}} = \frac{-2}{x}$$

Hole @ $x=-2$

$f(-2) = \frac{-2}{-2} = 1$

$\boxed{\text{Hole @ } (-2,1)}$



No x-int.

No y int (because $-\frac{2}{0}$ is U)

VA @ $x=0$

HA @ $y=0$, b/c bottom heavy

☆ To the right of the vertical asymptote we must find a point

Lets try $x=2$

$f(2) = \frac{-2}{2} = -1$

$\boxed{(2,-1) \text{ point on graph}}$

now sketch the branch